Word Ladder II (hard)

Problem Statement

A `transformation sequence` is a sequence that starts from the word `beginWord`, ends with `endWord`, and contains a word from the given dictionary `wordList` in the middle i.e. a sequence of words `beginWord -> s1 -> s2 -> ... -> sk` such that:

- Each adjacent pair of words differs by a single letter.
- Each s_i for `1 <= i <= k` is in wordList. sk == endWord
- The `beginWord` does not need to be in wordList.

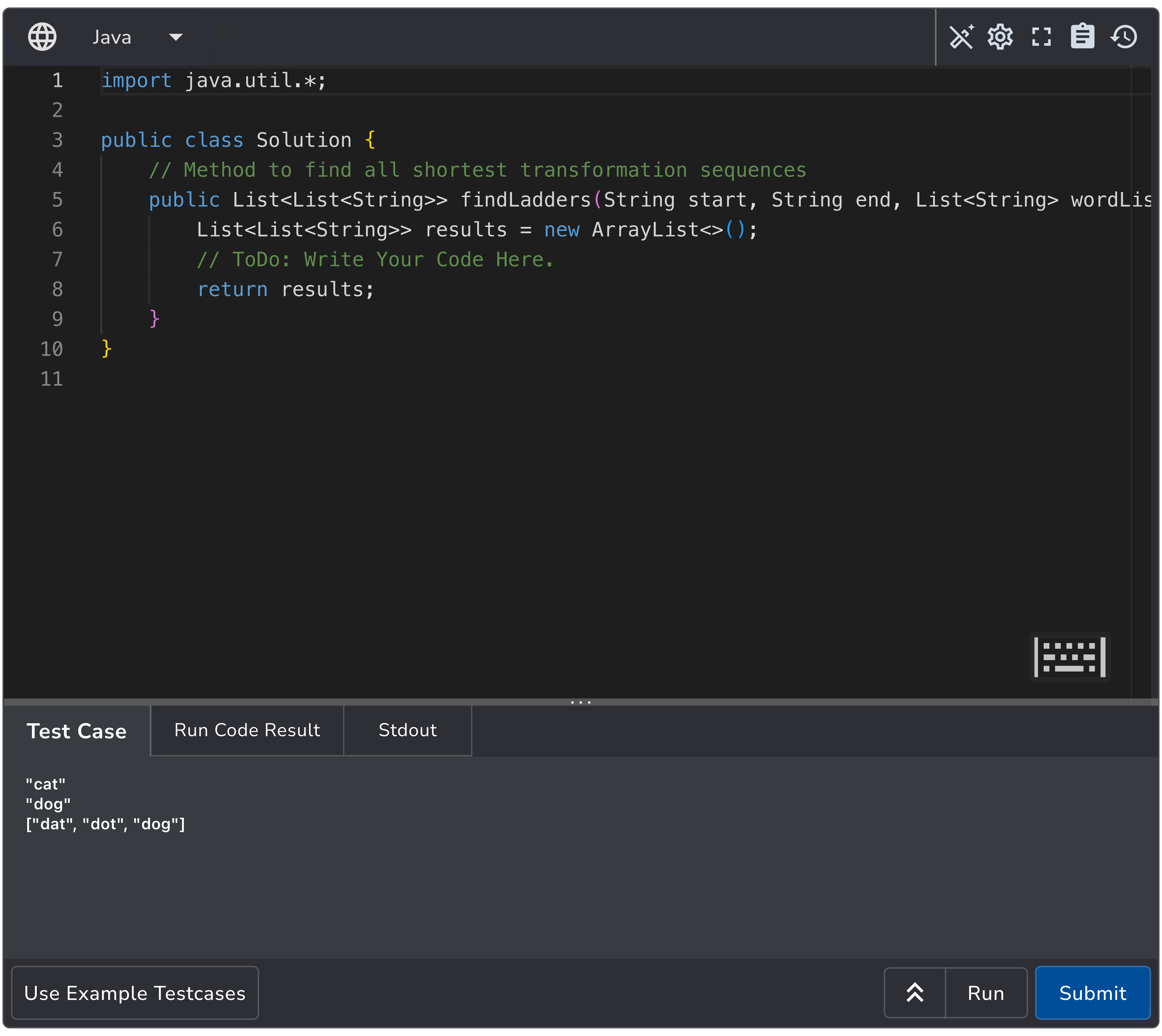
Given two strings, `beginWord` and `endWord`, and a `wordList`, return all the `shortest transformation sequences`, or an empty list if no such sequence exists. Each sequence should be in the form of `[beginWord, s1, s2, ..., sk]`.

Examples

1. **Example 1**
    - **Input:** start = "cat", end = "dog", wordList = ["dat", "dot", "dog"]
    - **Expected Output:** [["cat", "dat", "dot", "dog"]]
    - **Justification:** The sequence "cat" -> "dat" -> "dot" -> "dog" changes one letter at a time, and each word exists in the wordList. It's the shortest path from "cat" to "dog".
2. **Example 2**
    - **Input:** start = "bit", end = "fog", wordList = ["but", "put", "big", "pot", "pog", "dog"]
    - **Expected Output:** []
    - **Justification:** Although there are words in the list that are one letter away from "bit", there's no sequence that successfully transforms "bit" to "fog" through words in the wordList.
3. **Example 3**
    - **Input:** start = "hot", end = "cog", wordList = ["dot", "dog", "lot", "log", "cog"]
    - **Expected Output:** [["hot", "dot", "dog", "cog"], ["hot", "lot", "log", "cog"]]
    - **Justification:** This example shows two possible shortest transformation sequences from "hot" to "cog". Each sequence involves changing one letter at a time with all intermediate words existing in the given wordList.

Try it yourself

Try solving this question here:



```java
import java.util.*;

public class Solution {
    // Method to find all shortest transformation sequences
    public List<List<String>> findLadders(String start, String end, List<String> wordLis
        List<List<String>> results = new ArrayList<>();
        // ToDo: Write Your Code Here.
        return results;
    }
}
```

```
"cat"
"dog"
["dat", "dot", "dog"]
```

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Solution: Word Ladder

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Solution: Word Ladder II

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